

Numerical Optimization of Large-Scale Fiber-Optic Splitting Tree Uniformity

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Abstract—In order to split the optical power of a signal into many fibered outputs, large-scale optical splitting trees (splitters) are used. In most applications, the optical power should remain constant from output to output. Large-scale splitting trees (typically 1:32 or 1:128) may be composed of a single-stage or of cascaded small splitters (typically 1:2). In the former case, power uniformity is determined by the design and manufacturing tolerances. In the latter case, considered here, uniformity may be optimized by a proper choice of the positions of the small splitters in the tree. Simulations show that less than 1-dB uniformity may be reached.

Index Terms—Distortion, laser fusion, optical communication, optical fiber polarization.

I. INTRODUCTION

LARGE-SCALE splitting trees (also called splitters or couplers depending on the direction in which they are used) are required for many low-cost optical network concepts or for broadcasting a signal [1], [2].

The laser megajoule (LMJ) also needs fibered large-scale splitting trees [3]. Its 300-m-long building is under construction near Bordeaux (France) and will deliver 1.8 MJ on a few millimeters deuterium–tritium capsule in a few nanoseconds to make it fuse. Only one to four fibered laser sources will feed all the 120 free-space preamplifier modules. Each preamplifier module will then feed two laser chains, making 240 beams focalizing synchronously on the target. Large-scale fibered splitting trees are required at the working wavelength of 1053 nm to split the signal of the sources. The polarization has to be maintained in order to take advantage of zero loss at Brewster angle in the chains. Large-scale splitting trees may be composed of a single stage or of cascaded 1:2 splitters (also called 3-dB splitters). However, no such single-stage polarization-maintaining splitting tree has been developed at this wavelength. Hence, cascaded 1:2 splitters should be used.

Uniformity, U defined here as the ratio between the maximum and the minimum output powers, is a key feature of splitting trees [2], [4], [5]. For one-stage trees, U only depends on the design and on the fabrication tolerances. However, for cascaded 1:2 splitters, U both depends on the features of the 1:2 splitters and on their positions in the tree. Provided that their individual splitting ratios are known, their positions may be optimized. This is the topic of this letter.

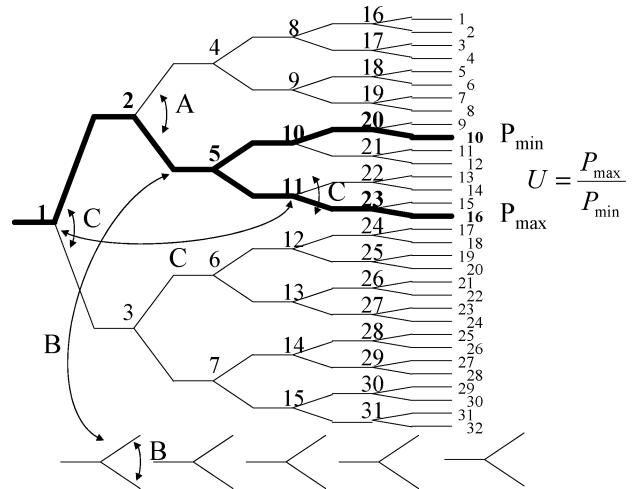


Fig. 1. Large-scale (here 1:32) splitting tree composed of many (here 31) 1:2 splitters. The bold lines define the critical path that leads to the output having the lowest and the highest optical powers (here Output 10 and 16 as an example). The arrows define the algorithm explained in the text.

In the following, we first compute the uniformity of a large-scale splitting tree with random placing of the 1:2 splitters. We then describe our algorithm to optimize their positions in a few minutes. We finally extend our algorithm to the optimization of several trees at the same time and show how to account for the splitter wavelength dependence if required.

II. RANDOM PLACING

A. Assumption

In order to build one 1: N splitting tree, $p = N - 1$ positions of the 1:2 splitters should be filled (see Fig. 1).

Suppose that $N_c(> p)$ 1:2 splitters are available; there are $N_c!/(N_c - p)!$ solutions to place the 1:2 splitters. For example, with $N = 128$ and $N_c = 135$, there are 10^{226} solutions! It is impossible to test all these solutions to find the best uniformity, even numerically before fabricating the tree. We have simulated the random placing of 1:2 splitters in a tree.

The optical power transmission of the arms of the 1:2 splitters is randomly drawn following a uniform distribution between 40% and 50%, which is realistic when we refer to our measurements. We neither consider the measurement accuracy nor the splicing loss standard deviation (~ 0.05 dB each) since they are small as compared to most uniformity values of the 1:2 splitters although they will slightly degrade the performance at the end. Fig. 2 gives the transmission of one arm versus the other of the drawing of $N_c = 135$ 1:2 splitters.

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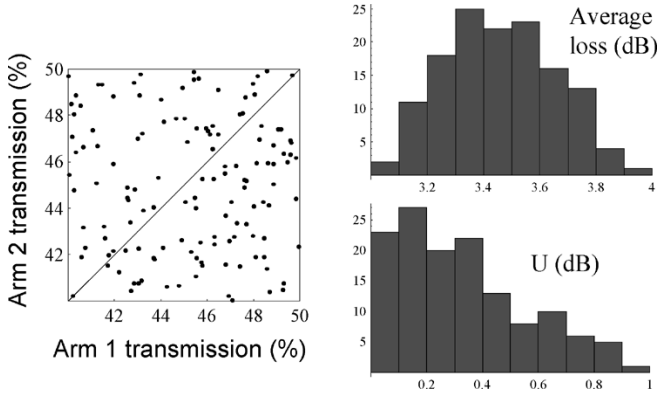


Fig. 2. Optical power transmission of one arm of $N_c = 135$ splitters versus the other assuming a uniform random distribution of power transmission between 40% and 50% (above). Deduced average loss histogram and uniformity of the 1:2 splitters (below).

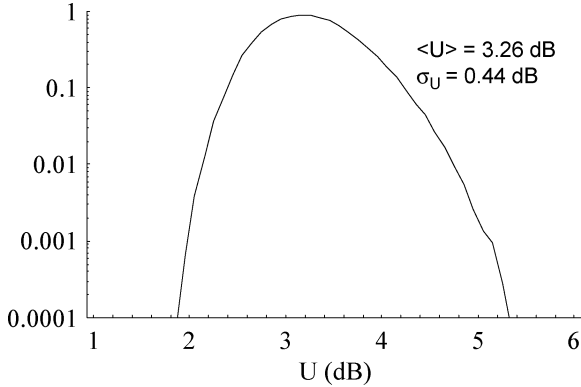


Fig. 3. PDF of U (in decibels) in log scale of the 1:128 splitting trees randomly composed of $p = 127$ 1:2 splitters out of the $N_c = 135$.

B. Simulation Results With Random Placing

Fig. 3 shows the probability density function (PDF) of U of a 1:128 splitting tree obtained with 300 000 random placing of $p = 127$ out of the $N_c = 135$ 1:2 splitters. Few 100 000 trees may be built per hour with a standard personal computer.

Random placing allows finding a configuration with 2-dB uniformity within a reasonable time. However, the probability to find a configuration with U below 1.5 dB within a few hours is almost zero with random placing. Nevertheless, as we shall see, among the $N_c!/(N_c - p)!$ solutions, lower values of U exist that should be found with a proper choice of algorithm.

III. ALGORITHM

The algorithm is sketched in Fig. 1. Starting from one random placing of a tree, it consists in modifying the critical path, defined as the 1:2 splitters in the tree involved in obtaining U . For instance, in Fig. 1, where $P_{\min}$ is obtained at the 10th output and $P_{\max}$ at the 16th, changing one of the 1:2 splitters number 1, 5, 10, 20, 11, or 23 modifies the powers at the 10th and 16th outputs, and thus, $P_{\min}$, $P_{\max}$, and U . Moreover, it may also change the positions of $P_{\min}$ and $P_{\max}$ in the tree, and thus, the critical path. Three kinds of changes are considered:

- A) Inverting one splitter. This is equivalent to interchange its branches.

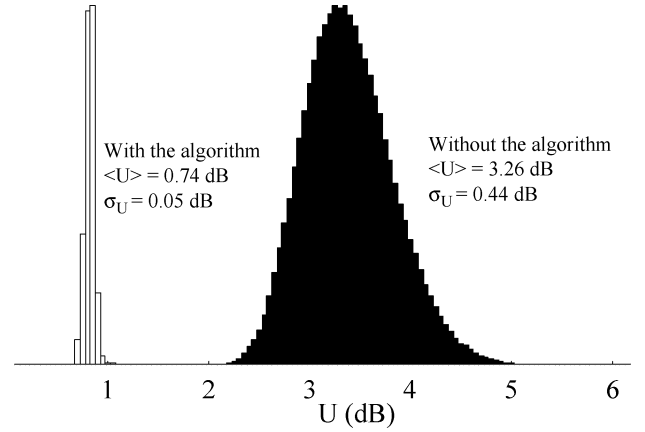


Fig. 4. PDF of U (in decibels) in linear scale of the 1:128 splitting trees randomly composed of $p = 127$ out of the $N_c = 135$ 1:2 splitters.

- B) Replacing one splitter of the critical path by one of the $N_c - p$ free splitters.

- C) Interchanging two splitters of the critical path.

For B and C, the different orientations of the 1:2 splitters should also be considered. The lowest value of U is chosen between all changes A, B, and C and the new critical path is computed as long as U decreases.

Depending on the initial random drawing of the tree, the obtained uniformity varies. The algorithm takes much more time than a simple random drawing but the uniformity should be much better.

We have applied this algorithm to find a configuration of the 1:128 tree to be build out of the $N_c = 135$ 1:2 splitters defined in Fig. 2. Fig. 4 shows the histograms of U in linear scale without (same as Fig. 3) and with the algorithm. A linear scale has been used since the algorithm is 15 000 slower than random placing. However, the probability to obtain U below 1 dB is almost equal to one with our algorithm. As a matter of fact, the maximum of U with the algorithm obtained during 15 h was 0.95 dB while the minimum value of U without the algorithm was 1.80 dB. Note that even without the algorithm, very low values of U may be reached as well. However, the algorithm dramatically increases the probability to get low values.

Only a few drawings with the algorithm are required to obtain values of U below 0.7 dB. It takes only a few minutes even with a personal computer and an interpreted mathematical language. Finding even lower values is still possible. As a matter of fact, the minimum value that was obtained during 15 h is 0.6 dB. However, measurement accuracies and splicing loss standard deviations will become predominant for very low values. Anyway, note that U cannot reach 0 dB even theoretically because U is higher than the uniformities of the $N/2 = 64$ 1:2 splitters at the last branches leading to the leaves.

The complexity of the algorithm increases with the total number of changes A, B, and C at each step, respectively, approximately proportional to $\log_2(N)$, $\log_2(N) \cdot (N_c - p)$, and $\log_2(N) \cdot (\log_2(N) - 1)$. Hence, as long as $N_c - p$, the number of free 1:2 splitters, remains low, the algorithm takes a reasonable time. When $N_c - p$ is high, the algorithm may be refined by sorting the 1:2 splitters into categories of splitters having similar features (similar average losses and similar

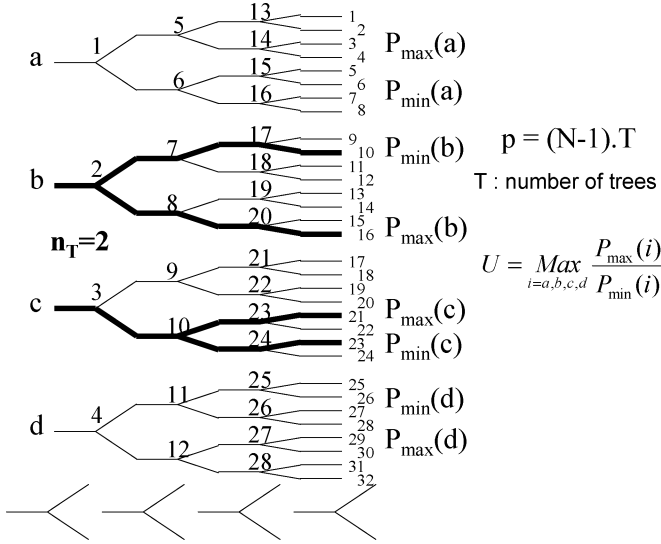


Fig. 5. Extension of the algorithm in the case of simultaneous optimization of T 1: N trees with N_c 1:2 splitters.

uniformities). Hence, the number of free splitters is replaced by the number of categories. This is equivalent to a reduction in measurement accuracy.

IV. EXTENSIONS

If one wants to optimize not one tree but T (>1) trees, the best way is to optimize them at the same time if all N_c 1:2 splitters are available. We consider that the overall uniformity of the trees is the worst (highest) uniformity among all the T trees. The algorithm sketched in Fig. 5 is modified as follows in order to allow change of splitters between two trees:

- 1) Set n_T to 1 (n_T : number of trees used to define the critical paths).
- 2) Find the n_T worst uniformities among the T trees.
- 3) Determine their critical paths.
- 4) Apply the aforementioned algorithm defined for a single tree on the splitters just defined.
- 5) When U cannot be decreased, increment n_T .
- 6) Stop when $n_T = T$.

We have applied this extended algorithm to improve the uniformity of four 1:32 trees with the same $N_c = 135$ 1:2 splitters defined in Fig. 2 ($T = 4$, $N = 32$ and $p = (N - 1) \cdot T = 124$). Results are shown in Fig. 6. The algorithm speed is almost equal to the optimization of a single tree and the best uniformity is also much below 1 dB.

Up to now, we have considered the case of a single wavelength, which is valid for LMJ. However, for a wavelength-di-

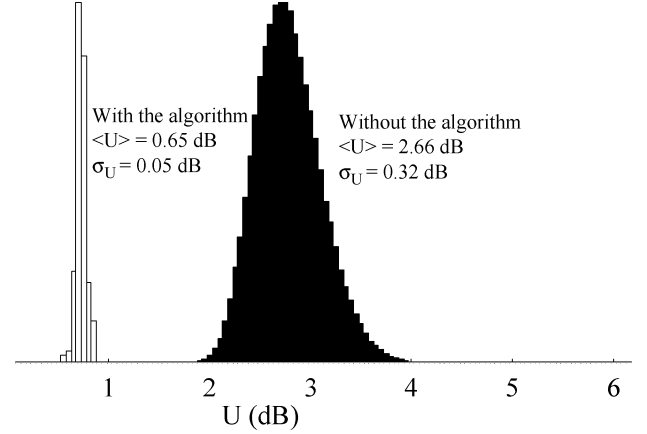


Fig. 6. PDF of U (in decibels) in linear scale of the four 1:32 splitting trees randomly composed of $p = 124$ out of the $N_c = 135$ 1:2 splitters.

vision-multiplexed (WDM) system, wavelength variation of the splitting ratio should also be considered. The algorithm may remain the same although the optimization criterion (U) should change. For instance, depending on the system requirements, one may either want to optimize either the spectral flatness of all outputs without consideration of the power variation from output to output or the power of all wavelengths of all outputs together even if it entails a degraded spectral flatness for some outputs.

V. CONCLUSION

We have shown an algorithm able to strongly reduce the power discrepancies in splitting trees composed of 1:2 splitters. When uniformity is key, it becomes easier to cascade 1:2 splitters than to tighten the specifications of large scale one-stage splitters. This algorithm may also be easily adapted to coupling ratio different from 50/50, to trees composed of cascaded 1: n ($n \neq 2$) splitters, or to WDM configurations.

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